

clock at the same time, so that we could see where the ball was at an exact time. This setup allowed us to make a lot of measurements. After several months of work, we determined that we could get a huge 44 percent edge in roulette by predicting the most likely octant in which the ball would land. So we built the first wearable computer, which is now in the MIT Museum.

How big was it?

It was about the size of a pack of cigarettes. The plan was the same one I had formulated sitting around the MIT co-op dining room one Sunday afternoon. One person would wear the computer and do timing with switches in the toes of his shoe. The other person would wear a radio receiver and make the bets.

I understand conceptually that Newtonian physics could be applied to predict a most probable landing area for the ball in roulette, but what I've never understood is how could you physically time a moving ball's location accurately enough for a computer to yield a usable answer.

That's a good question. The way it worked was we would use the hotel insignia on the outside stationary border of the wheel as a reference point, and each time the double zero passed this point, we would hit the switch to mark that time.

But how could you time that accurately enough?

That is where the strobe came in. We practiced in Shannon's basement with very subdued light, and the strobe would light up to show us where the ball actually was compared to where we thought it was. It took some training. We learned to anticipate just the right amount. With practice, we were able to get the standard error to within the ball's diameter.

I assume you have to wait for the ball to spin around a few times, so the velocity slows, before you make your estimate.

Yes, you basically want the ball to be going as slowly as possible when you place your bet, because the nearer you time it to the end, the more accurate your prediction.